



LR-N21-0062
13 August 2021

10CFR50.73

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-001

Hope Creek Generating Station
Renewed Facility Operating License No. NPF-57
Docket No. 50-354

Subject: Licensee Event Report 2021-001-00,
Safety Relief Valve (SRV) As-Found Setpoint Failures

In accordance with 10 CFR 50.73(a)(2)(i)(B), PSEG Nuclear LLC is submitting Licensee Event Report (LER) Number 2021-001-00, "Safety Relief Valve (SRV) As-Found Setpoint Failures."

There are no regulatory commitments contained in this letter.

Should you have any questions regarding this submittal, please contact Mr. Francis D. Possessky at (856) 339-1160.

Sincerely,

A handwritten signature in black ink, appearing to read "S. R. Poorman", with a long horizontal flourish extending to the right.

Steven R. Poorman
Plant Manager
Hope Creek Generating Station

Attachment: Licensee Event Report 2020-001-00

cc: Regional Administrator – Region I, NRC
US NRC NRR Project Manager
US NRC Senior Resident Inspector – Hope Creek
Manager NJ Bureau of Nuclear Engineering
Corporate Commitment Coordinator, PSEG Nuclear LLC



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: aira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Hope Creek Generating Station

2. Docket Number

05000354

3. Page

1 OF 3

4. Title

Safety Relief Valve (SRV) As-Found Setpoint Failures

5. Event Date

6. LER Number

7. Report Date

8. Other Facilities Involved

Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
06	14	2021	2021	-001	- 00	08	13	2021	Facility Name	05000
									Facility Name	Docket Number
										05000

9. Operating Mode

1 – Power Operation

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☐ 10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	☐ 10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	☐ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Francis D. Possessky, Regulatory Compliance Specialist

Phone Number (Include area code)

856-339-1160

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	SB	RV	T020	Y					

14. Supplemental Report Expected

15. Expected Submission Date

Month Day Year

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On June 14, 2021, Hope Creek Generating Station (HCGS) received results that two 'as-found' setpoint tests for safety relief valve (SRV) pilot stage assemblies (A and J) had exceeded the lift setting tolerance of +/- 3% of the nominal setpoint value prescribed in Technical Specification (TS) 3.4.2.1. Additionally, one SRV (R) did not lift when tested.

This is a condition reportable under 10 CFR 50.73(a)(2)(i)(B).

Setpoint exceedance for two of the SRV pilot stage assemblies is attributed to corrosion bonding between the pilot discs and seating surfaces, which is consistent with industry experience. This conclusion is based on previous cause evaluations and the repetitive nature of this condition at HCGS and within the BWR industry. The cause of the R SRV not lifting was main disc and piston thread wear. This conclusion is based on a failure analysis performed at the vendor.

The affected SRV pilot assemblies were replaced with pre-tested, certified spare assemblies. The R SRV was replaced.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Hope Creek Generating Station	05000- 354	2021	- 001	- 00

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric – Boiling Water Reactor (BWR/4)

Main Steam – EISS Identifier {SB}*

Safety Relief Valves – EISS Identifier {SB/RV}*

*Energy Industry Identification System {EISS} codes and component function identifier codes appear as {SS/CCC}

IDENTIFICATION OF OCCURRENCE

Event Date: June 14, 2021

Discovery Date: June 14, 2021

CONDITIONS PRIOR TO OCCURRENCE

When the reports of the 'as-found' results were received, Hope Creek was in Operational Condition (OPCON) 1, Operation at Power, at 100 percent rated thermal power. No other structures, systems or components that could have contributed to the event were inoperable at the time of the event.

DESCRIPTION OF OCCURRENCE

During the twenty-third refueling outage (H1RF23) at Hope Creek Generating Station (HCGS), seven 2-stage Main Steam safety relief valve (SRV) assemblies {SB/RV}, and one 3-stage SRV assembly were removed and tested at NWS Technologies. The SRVs are Target Rock Model 7567F two-stage SRVs and Target Rock Model 0867F 3-stage SRVs. On June 14, 2021, HCGS received the results of the 2-stage 'as-found' set pressure testing required by Technical Specification (TS) Surveillance Requirement (SR) 4.4.2.2. A total of two of the seven SRV pilot stage assemblies had setpoint drift outside of the required TS 3.4.2.1 tolerance values of +/-3% of nominal value.

The 'as-found' test results for the two 2-stage SRVs not meeting the TS requirements are as follows:

Valve ID	As Found (psig)	TS Lift Setting (psig)	Acceptable Band (psig)	% Difference Actual
F013A	1167	1130	1096.1 – 1163.9	3.27%
F013J	1161	1120	1086.4 – 1153.6	3.66%

During SRV Main Stage assembly testing required by TS SR 4.4.2.3 at NWS Technologies it was identified that the R SRV main valve did not lift.

Technical Specification (TS) 3.4.2.1 requires that the safety function of at least 13 of 14 SRVs be operable with a specified code safety valve function lift setting, within a tolerance of +/- 3%. Action (a) of TS 3.4.2.1 specifies "With the safety valve function of two or more of the above listed fourteen safety/relief valves inoperable, be in at least HOT SHUTDOWN within 12 hours and in COLD SHUTDOWN within the next 24 hours." Therefore, this is a condition reportable under 10 CFR 50.73(a)(2)(i)(B).



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Hope Creek Generating Station	05000- 354	2021	- 001	- 00

NARRATIVE

CAUSE OF EVENT

Setpoint exceedance for two of the SRV pilot stage assemblies (A and J) is attributed to corrosion bonding between the pilot discs and seating surfaces, which is consistent with industry experience. This conclusion is based on previous cause evaluations, the second lift test being within the acceptable band and the repetitive nature of this condition at HCGS and within the BWR industry.

The cause of the R SRV Main failing to actuate is attributed to the main disc and piston thread wear, resulting in side-loading of the main piston within the guide. This allowed slight grooves to form in the guide in which the piston and rings could bind. This conclusion is based on a failure analysis performed at NWS Technologies.

SAFETY CONSEQUENCES AND IMPLICATIONS

There were no instances during operating cycle 23 that resulted in any of the fourteen SRVs being declared inoperable and there were no events during that cycle that required operation of the SRVs. The industry has recognized that corrosion bonding occurs during the operating cycle. Once an SRV lifts, the corrosion bond breaks and subsequent openings occur very close to the set point as demonstrated during testing. The deviations observed following H1RF23 testing of SRVs are bounded by margins which exist in current Hope Creek design analyses; thus, the aggregate effect of this condition has no safety significance.

The R SRV was installed in H1RF20 and without firm evidence of the time that the R SRV would not actuate, this LER assumes for a safety consequence review, that the R SRV was inoperable since it was installed in H1RF20. A review was performed of the additional SRVs that were inoperable in Cycle 21 (B, D, F, G, H, L, M, P – setpoint failures) and Cycle 22 (A, C, D, H, J, K- setpoint failures) and determined that the ASME overpressure protection and core thermal limits would not have been exceeded.

PREVIOUS EVENTS

Similar events occurred during the 2016 (H1RF20) 2018 (H1RF21) and 2020 (H1RF22) Hope Creek refueling outages when multiple SRVs were found out of the TS required limits of +/- 3%. These events were reported as LER 354/2016-003-00, LER 354/2018-002-01 and LER 354/2019-002-00.

A three-year review was performed for other SRVs that failed to lift – no previous failures to lift were identified.

CORRECTIVE ACTIONS

1. All seven 2-stage SRV assemblies were removed and replaced with pre-tested, certified 3-stage SRV assemblies (H1RF23).
2. The R SRV was replaced with a 3-stage SRV model.

COMMITMENTS

There are no regulatory commitments contained in this LER.